Michael Grier

Software developer

Calgary, AB | michael.c.grier@gmail.com | 403-969-8220

GitHub: github.com/michael-grier | Website: michaelgrier.dev

I'm a full-stack developer with five years of experience building reporting and management applications and automating business processes. I focus on clean interfaces, reliable architecture, and clear communication. My background in education shapes how I explain technical decisions and collaborate with others.

Experience

Software Developer

TransAlta, Calgary | June 2023 to June 2026

Worked on a team of three software developers and three data engineers, sharing responsibility for architecture, UI design, development, testing, infrastructure, deployment, and maintenance.

Project: Athena | Production reporting and data governance application

- Contributed the majority of code to a Next.js and TypeScript production reporting platform with real-time dashboards for wind, hydro, solar, and natural gas management teams and senior leadership.
- Enabled SQL-fluent analysts to create their own dashboards within the platform using reusable ECharts cards and chart presets.
- Used Databricks as the primary analytics datastore, with PostgreSQL supporting latency-sensitive features such as authentication and comments.
- Built a data catalog management workflow with draft submissions, multi-stage review, rework, and audit trails.
- Implemented Microsoft Entra ID SSO, PostgreSQL-backed sessions, and distributed locking for token refresh across application instances.
- Built Terraform and GitHub Actions delivery workflows across development, UAT, and production, including release promotion, hotfixes, and rollback.
- Worked with analysts and management teams to define reporting requirements and refine dashboards through user feedback.

Project: GenOS | Wind, solar, hydro, and natural gas operations management application

- Refactored legacy React components across the application into smaller, focused modules to improve maintainability. Converted a hybrid REST/GraphQL architecture to pure GraphQL.
- Built comprehensive mobile and tablet views across the entire application, introducing responsive design where none existed before. Established Docker builds and Azure Pipelines across three environments.
- Fixed SQL Server connection leaks and strengthened JWT verification. Corrected capacity values, event filtering, and multi-unit CSV exports used in NERC reporting.

Junior Automation Developer

TransAlta, Calgary | June 2021 to May 2023

Developed and maintained automations using Blue Prism and Python scripting. Translated stakeholder requirements into reusable process components and handled testing, scheduled execution, and production troubleshooting.

Earlier experience

Freelance Software Developer

Self-employed, Calgary | February 2021 to June 2021

Led frontend design and development for a Django web project and supported Unity mobile game development.

Frontend Web Developer Intern

Invigulus, Remote | November 2020 to December 2020

Built a React application connected to a REST API and NoSQL database, giving employees a web interface for data entry and retrieval.

Selected projects

Independent development

Skateboard Ecommerce Platform

Checkout, inventory, and order management for an independent skateboard brand

Built checkout with server-validated pricing and atomic inventory reservations, retry-safe payment processing, and order management. Added end-to-end tests for price tampering, duplicate payments, and stock release.

Next.js, Stripe, Neon PostgreSQL, Drizzle, Cloudflare R2, Playwright

YYC Skate Spots

Community skateboarding map for Calgary

Built and released a mobile spot book on the App Store, with map clustering, location filters, photo submissions, favourites, moderation, and share links with a web fallback.

React Native, Expo, Convex, Clerk, Google Maps, EAS

Code Trainer

Interactive TypeScript interview practice

Built a browser-based coding workspace with isolated execution, TypeScript checking, and automated grading. Combined lessons and exercises with local guest progress and account-based synchronization across devices.

React, Vite, Monaco Editor, Web Workers, Convex, Better Auth, Vitest

Technical skills

Frontend and mobile: TypeScript, JavaScript, React, Next.js, React Native, Expo, Tailwind CSS, Material UI, shadcn/ui, ECharts

Backend and data: Node.js, Express, REST, GraphQL, PostgreSQL, SQL Server, Databricks, Convex, Drizzle, Zod, Python, FastAPI

Cloud and integrations: Azure, Terraform, Docker, GitHub Actions, Azure Pipelines, Stripe, Cloudflare R2, Microsoft Entra ID, Clerk, Better Auth

Testing and tools: Playwright, Vitest, Jest, React Testing Library, Bun, Blue Prism

Education

Object-Oriented Software Development Certificate | Southern Alberta Institute of Technology | 2020 · GPA 3.98

Bachelor of Education | University of Calgary | 2016

Bachelor of Arts | University of Calgary | 2011